

SWIR labeling logic for active-fire masks · conceptual band rule

(a) Distributed input bands (20 m)

B8A · 0.865 μm
NIR reference

B11 · 1.61 μm
SWIR context

B12 · 2.19 μm
hot-target response

Rendered PNG channels: R = B12, G = B11, B = B8A (fixed stretch; not auto-leveled)

(b) Modified HOTMAP seed and growth (Eqs. 2–4)

Seed C
 $B12 \geq 0.35$ and $B12/B8A \geq 4$

Potential P
 $B12 \geq 0.25$ and $B12/B8A \geq 2$

Grown mask
2× 4-neighbor growth of C into P

Confirmed seeds mark flaming cores. Cooler front margins enter the mask only when they touch a seed through two constrained growth steps. Isolated potential pixels are rejected. Diagram is conceptual — not measured spectra.